

Public Cloud Services IaC – Configuration Mgmt



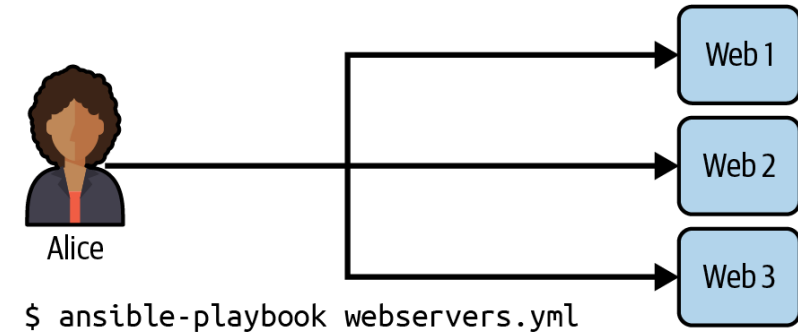
Use Case: Administrating multiple server

Task:

- Install NGINX
- Generate NGINX-config
- Restart Nging
- ...and this on three servers...

Ansible:

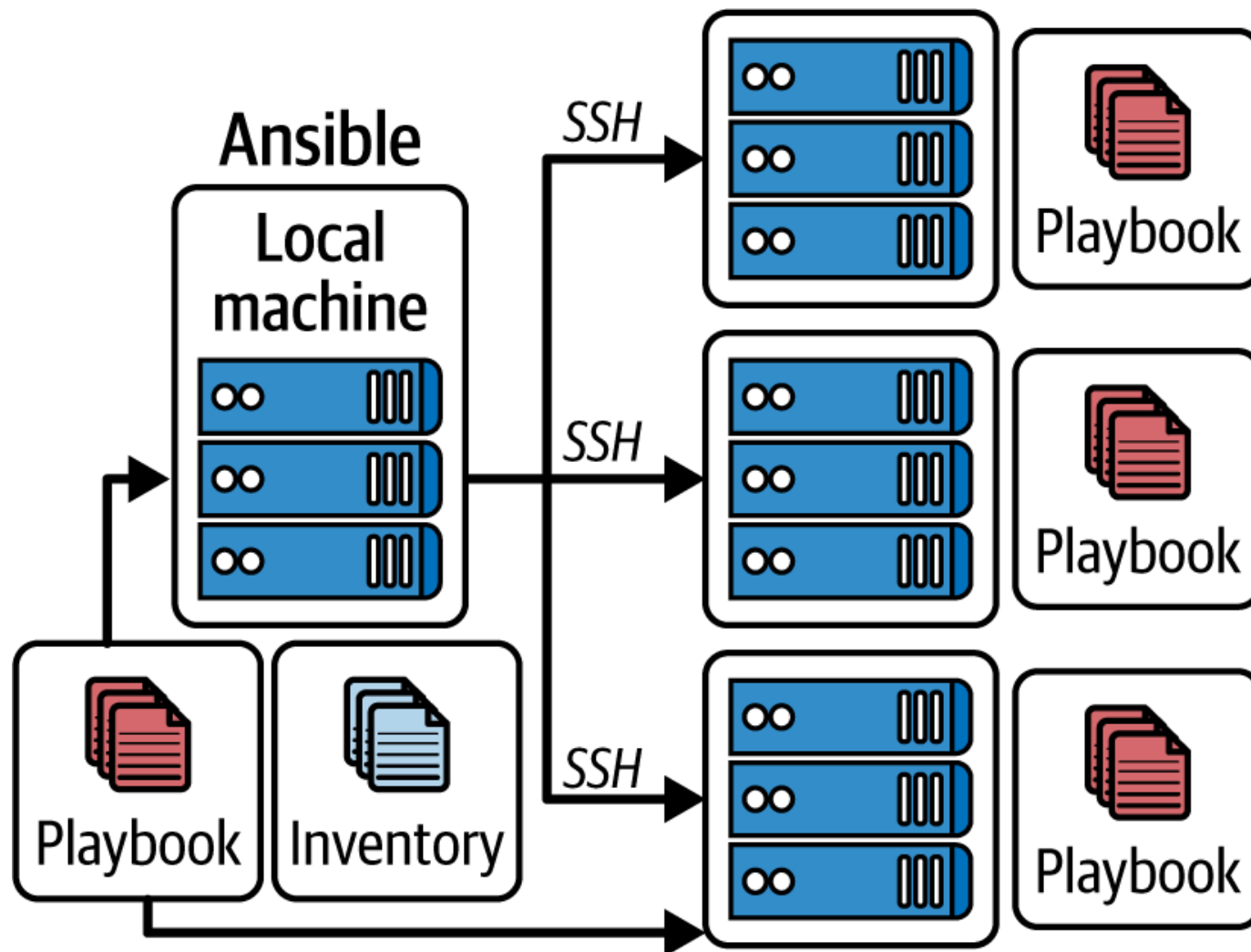
- Idempotence (if done right)
- No extra agent, just SSH
- No extra state (gathering facts before)



Playbook: webservers.yml

```
---
- name: Configure webservers
  hosts: webservers
  become: True
  tasks:
    - name: Install nginx
      package: name=nginx
    - name: Install config file
      template:
        src: nginx.config.j2
        dest: /etc/nginx/nginx.conf
      notify: Restart nginx
  handlers:
    - name: Restart nginx
      service: name=nginx state=restarted
```

Architecture



<https://learning.oreilly.com/library/view/red-hat-certified/9781098162122/ch02.html>

Inventory

- Describes all servers to be orchestrated
- Might be distinguished into groups
- Can be dynamic (if executed against cloud):
Inventory is a script in this case

<https://docs.ansible.com/ansible/latest/plugins/inventory.html#id3>

```
[webservers]
192.168.1.92
192.168.1.94
192.168.1.96

[dbservers]
192.168.1.93
```

```
! openstack.yaml ×
inventory > ! openstack.yaml > [abc] fail_on_errors
1   plugin: openstack.cloud.openstack
2   expand_hostvars: yes
3   fail_on_errors: yes
4   all_projects: no
```

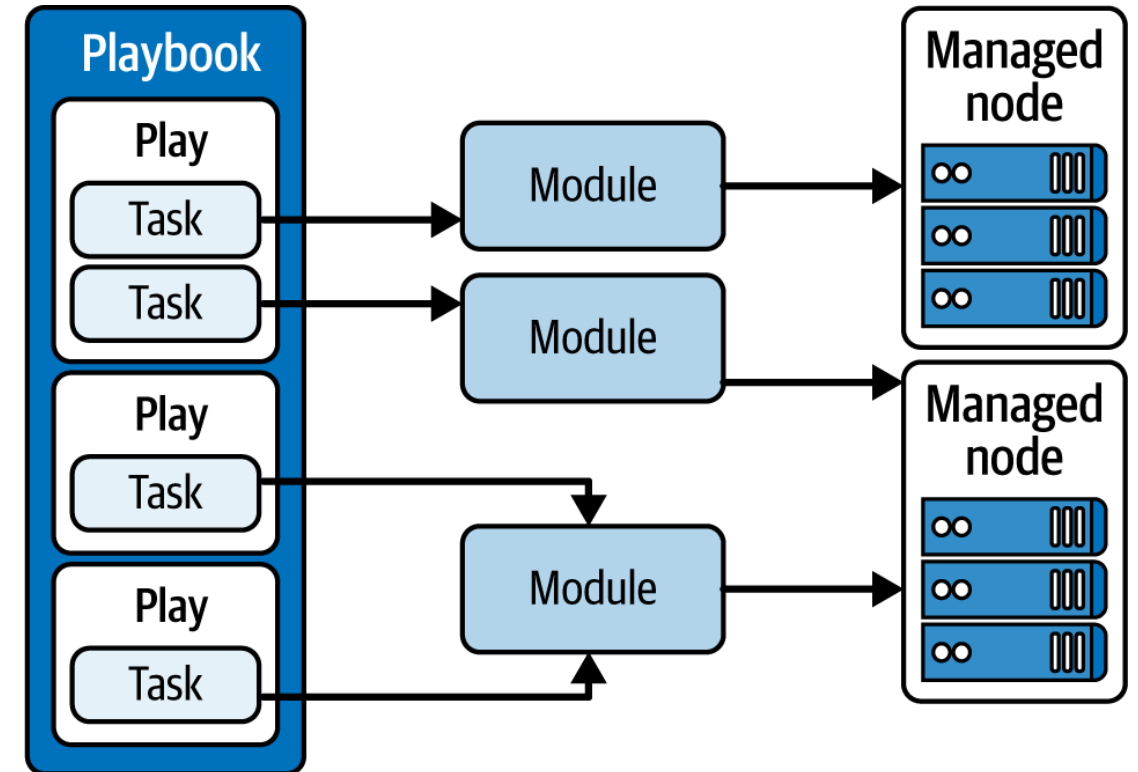
Playbook

```
---
- name: Web Servers 1
  hosts: webservers 2
  become: true 3

  tasks: 4
  - name: Ensure apache is at the latest version
    ansible.builtin.yum: 5
      name: httpd 6
      state: latest 7

- name: DB servers
  hosts: databases
  become: true

  tasks:
  - name: Ensure postgresql is at the latest version
    ansible.builtin.yum:
      name: postgresql
      state: latest
  - name: Ensure that postgresql is started
    ansible.builtin.service: 8
      name: postgresql
      state: started 9
```



<https://learning.oreilly.com/library/view/red-hat-certified/9781098162122/ch03.html>

Puppet

- Consists of Puppet-Master and Puppet-Slave
- Needs Config-Repository
- Needs
 - dedicated packages on Agent and Master
 - Open high ports
 - Systemd-Services
- No inventory needed since register-approach

```
# Für Webserver
node 'webservers' {
  package { 'httpd':
    ensure => latest,
  }

  service { 'httpd':
    ensure    => running,
    enable    => true,
    subscribe => Package['httpd'],
  }
}

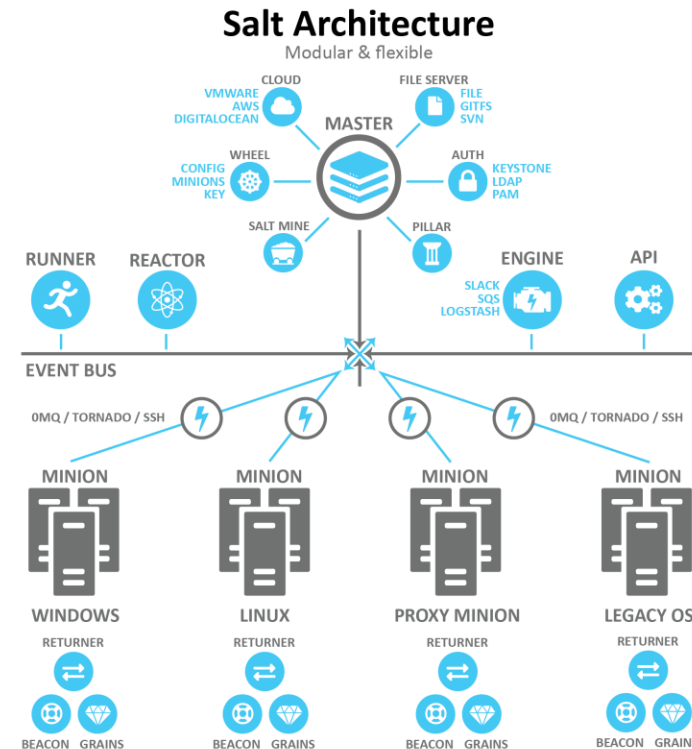
# Für Datenbankserver
node 'databases' {
  package { 'postgresql':
    ensure => latest,
  }

  service { 'postgresql':
    ensure    => running,
    enable    => true,
    subscribe => Package['postgresql'],
  }
}
```


SaltStack

Feature-complete version for infra and conf-mgmt:

- can also provision infra with multiple cloud bindings
- Performs conf-mgmt. with own protocol and master/minion-setup
 - Fallback to ssh possible but slower
 - Event-triggering mechanism in case of changes
 - Minion phones to master



```
# top.sls
base:
  'webservers':
    - apache
  'databases':
    - postgresql
```

```
# postgresql.sls
postgresql:
  pkg.installed:
    - name: postgresql
  service.running:
    - name: postgresql
    - enable: true
    - require:
      - pkg: postgresql
```

```
# apache.sls
apache:
  pkg.installed:
    - name: httpd
  service.running:
    - name: httpd
    - enable: true
    - require:
      - pkg: apache
```

